IoTPentBench: A Network-Scale Benchmark for LLM-Based IoT Penetration Testing

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***Abstract*—Autonomous LLM penetration-testing agents are typically evaluated on single hosts, yet IoT deployments expose risk through network reachability: gateways, brokers, cameras, and OT services form attack paths whose security depends on pivots across devices. We present IoTPentBench, a reproducible network-scale IoT benchmark with 15 Proxmox/Ansible scenarios, 145 devices, and 229 injected vulnerabilities mapped to OWASP IoT Top 10 and MITRE ATT&CK for ICS. We also introduce MESHSCOUT, a six-phase network-native harness that combines topology priors, active discovery, per-device triage, per-vulnerability verification, intrusion attempts, and report generation. The benchmark adds two metrics absent from single-host evaluations: evidence-level coverage (L1 detection, L2 exploitation, L3 exfiltration) and Multi-Hop Reach (MHR$_k$), which measures recall on vulnerabilities that require at least k pivots or zone transitions. We describe the benchmark, evaluator, and experimental protocol, and use them to isolate architectural effects independently from model choice.**

***Index Terms*—IoT security, penetration testing, large language models, autonomous agents, benchmark, cyber-physical systems**

1. Introduction

Modern IoT deployments are *networks*, not hosts: a smart building chains cameras, brokers, and controllers behind a gateway; an industrial site puts Modbus PLCs behind a jump host. The attack surface that matters is the union of reachability paths, and real-world incidents (Mirai [25], Verkada [26], Stuxnet [27]) all materialised as multi-hop pivots across that surface.

LLM-pentesting agents have matured rapidly and report substantial gains on public test corpora (Table 2), yet every such corpus is a collection of *isolated containers or VMs* and every agent is scored one target at a time. Three properties of a real IoT deployment remain invisible to that metric: *hop depth* (behind-pivot vulnerabilities), *IoT/OT protocols* (MQTT, Modbus, LoRaWAN, CoAP), and *evidence depth* (detection vs exploitation vs exfiltration). These gaps are properties of the evaluation surface, not of the agent: improving the agent alone does not close them.

Contributions.

- **IoTPentBench**: the first reproducible network-scale benchmark for LLM pentest agents on IoT. It ships 15 scenarios (10 main covering 5 architectural patterns, 3 scalability stressors up to 35 devices, 2 hardened controls for False Positive rate), 229 vulnerabilities injected via Ansible, with ground truth mapped to OWASP IoT Top 10 and MITRE ATT&CK for ICS.
- **MESHSCOUT**: a network-native harness whose six phases (topology prior, discovery, per-device triage, per-vulnerability verification, intrusion, report) keep LLM context tractable at $/24$ scale.
- **L1/L2/L3 evidence scoring and MHR$_k$**: metrics that separate detection, exploitation, and exfiltration, and that measure recall as a function of firewall-hop distance.
- **Evaluation protocol** holding the LLM constant across systems to isolate architectural contributions. The primary comparison uses MESHSCOUT, two internal ablations, and Nmap NSE; optional LLM-agent baselines are treated as adapters when their tool interfaces can be reproduced under the same model and budget.

All artifacts (Ansible playbooks, ground truth, evaluator, run logs) are released at `ANONYMIZED_URL`.

2. Related Work

This section reviews prior work relevant to our approach. We first survey LLM-driven penetration-testing agents and the benchmarks used to evaluate them, then discuss the classical attack-graph literature that informs MESHSCOUT's topology-aware design, and finally review IoT firmware analysis and the security frameworks underlying our scenario taxonomy.

2.1. LLM Penetration-Testing Agents

Happe and Cito [1] were among the first to show that a single LLM driving a shell can autonomously perform privilege-escalation tasks on Linux VMs. PENTESTGPT [3] then introduced a multi-module design (three coordinated LLM sessions over a *Pentesting Task Tree*), evaluated on 182 hand-curated HackTheBox and VulnHub sub-tasks across 13 targets, reaching 66.5% subtask and 46.2% end-to-end completion against 52.2%/38.5% for plain GPT-4 and

23.1%/7.7% for GPT-3.5. HACKINGBUDDYGPT [2], AUTOATTACKER [4] and NEBULA [5] extend this single-agent line into open-source tools.

Later systems push specialisation from cognitive roles to penetration phases. VULNBOT [6] decomposes the workflow into three phases (reconnaissance, scanning, exploitation) driven by five cooperating modules over a *Penetration Task Graph*; it reports 69.1% subtask and 30.3% end-to-end completion on AutoPenBench with Llama 3.1-405B, a 21-point end-to-end gain over the same model used single-agent. AUTOPENTESTER [7] adds retrieval-augmented generation across five LLM agents and reports a +27% subtask gain over PENTESTGPT on HackTheBox. PENTESTAGENT [8] and the production-grade CAI [9] generalise this multi-agent blueprint with modular tool use and a public leaderboard.

A complementary thread improves the underlying agent rather than its orchestration. PENTEST-R1 [10] applies two-stage reinforcement learning on reasoning traces, and xOFFENSE [11] LoRA-fine-tunes Qwen3-32B on offensive write-ups (TryHackMe, HackTheBox, VulnHub, WHITERABBITNEO), reaching 79.17% subtask completion on AutoPenBench and surpassing PENTESTGPT and VULNBOT. CHECKMATE [16] couples an LLM agent with a *Classical Planning+* module encoding preconditions and effects of predefined attack actions, improving success rates by more than 20% over Claude Code + Sonnet 4.5 on the Vulhub benchmark.

Despite the diversity of these designs, published scores are usually reported at the level of a single target or challenge (a HackTheBox machine, a VulnHub VM, an AutoPenBench task, or a CTF environment). Some environments require multi-step exploitation, but the reported metrics do not expose subnet-level reachability, hop depth, or per-vulnerability progress across an IoT/OT topology with protocols such as MQTT, Modbus, LoRaWAN, and CoAP.

2.2. LLM Cybersecurity Benchmarks

Challenge-level corpora. AUTOPENBENCH [17] is the dominant evaluation substrate — 24 in-vitro tasks plus 12 real-world CVE-based exploits with scripted flag checkers — and is the corpus behind VULNBOT, PENTEST-R1, and xOFFENSE; VULHUB [24] fills the same role for CHECKMATE with ~100 Docker images (one container per CVE). CAIBENCH [21] is the broadest academic effort, spanning CTFs, attack-and-defense ranges, and knowledge tests; its RCTF2 subset adds 27 robotics challenges — the first cyber-physical inclusion in this family. These corpora can require multi-step reasoning, and some challenge environments are richer than a single shell. However, their public scoring is challenge- or flag-level: they do not provide a per-vulnerability ground truth tied to explicit network reachability, hop depth, or IoT/OT application protocols (MQTT, Modbus, LoRaWAN). Table 2 reports headline results for each system.

Multi-host labs. LUDUS [22] and the GOAD [23] Active-Directory scenario deliver Proxmox+Ansible deployments of representative enterprise networks, the closest methodological precedent to our infrastructure. They do model multi-host compromise and lateral movement, so the gap is not the existence of multi-host cyber ranges. Rather, these labs are not packaged as LLM-agent benchmarks: they do not ship a per-vulnerability ground-truth schema, hop-depth annotations, or a scoring procedure, and they target enterprise AD rather than IoT/OT protocols. No prior benchmark therefore combines multi-host reachability, IoT/OT protocols, per-vulnerability machine-readable ground truth, and evidence-level scoring beyond binary flag capture (Table 1).

2.3. Attack Graphs and Network Reachability

Representing a network as a graph for security analysis predates deep learning by decades. Phillips and Swiler [28] introduced graph-based network-vulnerability analysis, Sheyner *et al.* [29] developed automated attack-graph generation via model checking, and MULVAL [30] cast the problem as Datalog inference, becoming the de facto reference for attack-graph research. These systems compute reachable attacker states from a *static* encoding of host configurations and exposed services. We borrow their reachability abstraction (nodes are devices, edges are firewall-permitted protocol flows) but MESHSCOUT operates on a *live* network: edges are validated by actual probes, the LLM proposes the next action, and the graph is updated as new hosts are discovered. Classical attack-graph generation is therefore complementary rather than competing.

2.4. IoT Security Context

Scenario design and ground-truth taxonomy follow three references: the OWASP IoT Top 10 [31] for vulnerability categories, MITRE ATT&CK for ICS [32] for OT-tactic annotations, and NIST SP 800-82 Rev. 3 [33] for segmentation conventions, notably the IT/OT split underpinning pattern P3 (§4.2). On the empirical side, the closest IoT security work targets firmware rather than deployed networks: Costin *et al.* [34] performed the first large-scale firmware study, FIRMADYNE [35] introduced dynamic firmware emulation for vulnerability discovery, and FIRMAE [36] substantially improved emulation success rates. OWASP IOTGOAT [37] provides a deliberately insecure firmware image used as a teaching artefact. These projects produce vulnerable images; IoTPentBench produces a deployed multi-device network with firewall-enforced reachability, the dimension along which we evaluate MESHSCOUT.

3. Threat Model and Problem Definition

Network-scale pentest. We define *network-scale penetration testing* as the task of: (i) discovering all live hosts in a target IP range, (ii) enumerating services and identifying vulnerabilities on each host, (iii) producing evidence of

TABLE 1. Comparison with prior LLM pentest systems and benchmarks. *Scope*: published unit of evaluation (target/challenge = one target, task, or flag environment; network = multi-device IP range with explicit reachability). *Domain*: protocols / target type. *Test corpus*: dataset on which results are reported. Rows are grouped by role: LLM pentest agents (top) and evaluation corpora (bottom).

System	Scope	Domain	Test corpus	IoT-specific	Scoring
LLM pentest agents					
Happe & Cito [1]	target/challenge	Web/OS	curated Linux VMs	no	flag
PentestGPT [3]	target/challenge	Web/OS	HTB + VulnHub (13 VMs)	no	sub-task
VulnBot [6]	target/challenge	Web/OS	AutoPenBench	no	task-graph
AutoPentester [7]	target/challenge	Web/OS	HackTheBox	no	sub-task
CAI [9]	target/challenge	Web/CTF	CAIBench	no	CTF
Pentest-R1 [10]	target/challenge	Web/OS	AutoPenBench	no	sub-task
xOffense [11]	target/challenge	Web/CTF	AutoPenBench	no	sub-task
EnIGMA [12]	target/challenge	CTF	NYU CTF Bench	no	flag
CheckMate [16]	target/challenge	Web/OS	Vulhub	no	sub-task
Evaluation corpora					
AutoPenBench [17]	target/challenge	Web/OS	24 in-vitro + 12 CVEs	no	flag
NYU CTF Bench [18]	target/challenge	CTF	200 Jeopardy challenges	no	flag
Cybench [19]	target/challenge	CTF	curated CTF subset	no	flag
CAIBench / RCTF2 [21]	target/challenge	mixed	meta-benchmark	27 robotics	CTF
Vulhub [24]	target/challenge	Web/OS	∼100+ Docker CVEs	no	custom
Ludus + GOAD [22], [23]	multi-host	AD/Win	AD lab (no GT)	no	–
IoTPentBench (ours)	**network**	**MQTT/Modbus/LoRa/CoAP/. . .**	**15 IoT scenarios (up to 35 dev.), per-vuln YAML**	**yes**	**L1/L2/L3 + MHR$_k$**

TABLE 2. Headline numbers reported by prior LLM pentest systems on their chosen corpus. *Sub-task %* = fraction of predefined intermediate steps solved. *E2E %* = fraction of full target/challenge solved end-to-end. Values are absolute except where prefixed with "+" (relative gain vs the immediately preceding baseline row). Rows are not directly comparable across corpora: the table makes explicit *where* each tool is tested and *what* number it publishes. No prior result is reported on a multi-device IoT network with firewall-enforced reachability, which is the gap IoTPentBench targets.

System	Model	Corpus	Sub-task %	E2E %
GPT-3.5 baseline [3]	GPT-3.5	HTB+VulnHub (13 VMs)	23.1	7.7
GPT-4 baseline [3]	GPT-4	HTB+VulnHub (13 VMs)	52.2	38.5
PentestGPT [3]	GPT-4	HTB+VulnHub (13 VMs)	**66.5**	**46.2**
AutoPentester [7]	GPT-4o	HackTheBox	**+27.0**†	n/r
Llama 3.1-405B baseline [6]	Llama 3.1-405B	AutoPenBench	49.1	9.1
GPT-4o baseline [6]	GPT-4o	AutoPenBench	n/r	21.2
VulnBot [6]	Llama 3.1-405B	AutoPenBench	**69.1**	**30.3**
xOffense [11]	Qwen3-32B (LoRA)	AutoPenBench	**79.2**	n/r
Claude Code (baseline) [16]	Sonnet 4.5	Vulhub	n/r	baseline
CheckMate [16]	Sonnet 4.5 + Plan+	Vulhub	n/r	**+20**†
EnIGMA [12]	GPT-4	NYU CTF (200)	n/r	SOTA
IoTPentBench (ours)	**shared (see §6)**	**15 IoT scenarios (up to /24, 35 dev.)**	**Recall / Prec / F1 / Score / MHR$_k$ / L1L2L3**	

† *relative gain over the immediately preceding baseline row. "n/r" = metric not reported in the source paper.*

exploitability where feasible, and (iv) aggregating findings at the network level. The *unit of evaluation* is the network, not an individual host.

Attacker model. The attacker holds network-layer access to the target subnet (post-ingress, pre-compromise). They hold no prior credentials, no privileged vantage, and no knowledge of deployed software beyond what is observable through scans. The attacker has access to standard offensive tooling (Nmap, MQTT clients, SSH, . . .) and an LLM with tool-calling. Time and budget are bounded.

Goals.

1) Enumerate all devices reachable from the entry point.
2) For each device, identify vulnerabilities from misconfiguration, weak credentials, information disclosure, insecure protocols, and known CVEs.
3) For each finding, attempt to produce *evidence* at one of three levels: L1 detection (port open, banner leak), L2 exploitation (authenticated session, protocol handshake), L3 data exfiltration (configuration files, credentials, sensor data).
4) Report findings at the network level (severity-ranked, deduplicated).

Scope and non-goals. We evaluate at the *IP/application layer* on deployed IoT networks. We do *not* evaluate: (a) physical/side-channel attacks on hardware, (b) RF-layer attacks requiring SDR captures, (c) firmware extraction from physical devices, (d) post-exploitation persistence or impact, (e) insider threats or supply-chain compromise. Hardware-assisted attacks are left as future work (§10).

4. IoTPentBench: A Network-Scale Benchmark

4.1. Design principles

IoTPentBench is built around six design principles derived from the gap analysis in §2:

- **Reproducibility.** The entire infrastructure deploys from Ansible playbooks against a Proxmox cluster; every vulnerability is injected idempotently with no manual post-deployment steps.
- **Architectural diversity.** Scenarios span five network patterns (§1) with distinct attacker reachability graphs encoded in the topology metadata and, for segmented/VLAN cases, enforced by OpenWrt firewall rules rather than service labels alone.
- **Protocol coverage.** Each scenario mixes multiple IoT and OT protocols (MQTT, Modbus, LoRaWAN gateways, CoAP, SNMP, plus standard SSH/HTTP/FTP/Telnet) to exercise protocol-specific reasoning.
- **Ground-truth granularity.** Every injected vulnerability is documented in YAML with `device`, `IP`, `role`, `severity`, `category`, `OWASP IoT` and `MITRE ICS` mappings, optional `CVE`, `indicators` (expected detection strings), and a `verification` command.
- **Scalability axis.** Three scenarios (s_11, s_12, s_13) stress network size at 15, 17, and 35 devices respectively, while the 10 main scenarios stay in the 4–8 device range. This separates the pattern-coverage axis from the size axis.
- **Hardened controls.** Two scenarios (s_1h, s_4h) reuse the topologies of s_1 and s_4 with all injected vulnerabilities removed. They measure agent False Positive rate on a deployment that should return "no findings".

4.2. Five architectural patterns

Concretely:

1) **P1 — Flat.** All devices are directly reachable from the entry point; no pivot is required.
2) **P2 — Gateway-centric.** A single IoT gateway mediates all access to back-end services; inter-service traffic is blocked without gateway compromise.
3) **P3 — Segmented IT/OT.** Three zones (admin/IT/OT) with access-control lists; the OT zone (Modbus PLCs, HMI) is reachable only via a jump host.
4) **P4 — Hub-centric star.** A central home-automation hub is the only device that can reach MQTT broker, DB, and camera; peripherals cannot talk to each other directly.
5) **P5 — Edge-cloud pivot.** Edge subnet and cloud subnet are separated; only the edge gateway bridges them, making cloud-side services unreachable without edge compromise.

TABLE 3. THE 15 SCENARIOS OF IOTPENTBENCH. *Main* SCENARIOS COVER THE 5 ARCHITECTURAL PATTERNS; *Scalability* SCENARIOS STRESS NETWORK SIZE; *Hardened* CONTROLS SHIP 0 VULNERABILITIES TO MEASURE FALSE POSITIVE RATE.

ID	Pattern / role	Dev.	Vulns	Theme
Main scenarios (pattern coverage)				
`s_1`	P1 Flat	4	12	Minimal IoT (MQTT/Web/SSH)
`s_2`	P1 Flat	6	13	Flat IoT + IT mix
`s_3`	P2 Gateway-centric	8	18	NATO lab replica
`s_4`	P3 Segmented IT/OT	8	18	Industrial (Modbus/HMI)
`s_5`	P4 Hub-centric star	8	15	Smart building (cameras/NVR)
`s_6`	P4 Hub-centric star	6	16	Home automation hub
`s_7`	P5 Edge-cloud pivot	6	14	Edge → cloud API
`s_8`	P3 Segmented IT/OT	8	14	IT/IoT/OT 3-zone industrial
`s_9`	P1 Flat (mesh)	6	11	Mesh IoT smart-city outdoor
`s_10`	P1 Flat (variants)	6	13	Role-variant stress
Scalability stressors				
`s_11`	Smart City (flat)	15	23	3 zones same /24 (no isolation)
`s_13`	VLAN-segmented	17	20	Real Proxmox VLAN bridge (3 VLANs)
`s_12`	Smart City (large)	35	42	Largest network (34 LXC services)
Hardened controls (FP rate)				
`s_1h`	P1 Flat (hardened)	4	0	s_1 with vulns removed
`s_4h`	P3 ICS (hardened)	8	0	s_4 with vulns removed
Total		**145**	**229**	

IoTPentBench ships 15 scenarios organised on three axes: 10 *main* scenarios that instantiate the 5 architectural patterns above (with small-to-medium device counts, 4–8 devices each), 3 *scalability* scenarios that stress network size (15, 17, and 35 devices) using logical multi-zone and real-VLAN variants, and 2 *hardened controls* that ship identical topologies to s_1 and s_4 *without* any injected vulnerability, used to measure agent False Positive rate. The full mapping is given in Table 3.

4.3. Deterministic vulnerability injection

Vulnerabilities are injected by a single Ansible role (`04_inject_vulns.yml`) parameterized by `scenario_id`. Each scenario YAML lists `router_vulns` (OpenWrt-level: Telnet, WAN admin, FTP) and per-service `role` definitions that trigger role-specific injections: *e.g.*, the `mqtt_broker` role sets `allow_anonymous true` and binds on `0.0.0.0:1883`; the `ssh_server` role installs weak credentials (`admin:admin`) and enables `PasswordAuthentication`; the `db_server` role starts MariaDB with a passwordless `root`. Category distribution is shown in §4.

4.4. Ground-truth schema

Each scenario ships a YAML file describing every injected vulnerability. An entry example (from `scenario_1.yaml`):

Fig. 1 placeholder: five topology diagrams side-by-side.
(Flat) — (Gateway-centric) — (Segmented IT/OT) — (Hub-star) — (Edge-cloud)

Figure 1. The five architectural patterns in IoTPentBench. Reachability is encoded in the scenario topology and enforced by OpenWrt firewall rules in the segmented/VLAN cases.

TABLE 4. VULNERABILITY CATEGORIES AND SEVERITY DISTRIBUTION ACROSS ALL 13 VULNERABILITY-INJECTED SCENARIOS OF IOTPENTBENCH (229 INJECTED INSTANCES; THE 2 HARDENED CONTROLS CONTRIBUTE 0 BY DESIGN). COUNTS DERIVED FROM `GROUND_TRUTH/SCENARIO_*.YAML`.

Category	Count	CRIT	HIGH	MED	LOW
`misconfiguration`	68	7	36	25	0
`no_authentication`	41	17	21	3	0
`data_exposure`	38	0	14	24	0
`default_credentials`	29	3	26	0	0
`info_disclosure`	13	0	0	0	13
`code_injection`	11	10	1	0	0
`cve`	9	0	3	6	0
`insecure_update`	9	0	4	5	0
`weak_crypto`	6	0	0	1	5
`missing_header`	3	0	0	0	3
`privilege_escalation`	2	0	2	0	0
Total	**229**	37	107	64	21

```
- id: V1
  device: s1-mqtt
  ip: "192.168.100.11"
  role: mqtt_broker
  title: "MQTT without authentication"
  severity: high
  category: misconfiguration
  owasp_iot: "I1 / I9"
  mitre_ics: "Initial Access, Collection"
  cve: null
  indicators:
    - "allow_anonymous true"
    - "Anonymous clients connected"
  verification: "mosquitto_sub -h 192.168.100.11 -t '#' -v"
```

Figure 2. Ground-truth entry example.

Each scenario YAML may additionally declare a top-level `bonus_types` list of vulnerability types that are *adjacent* to the injected set (e.g., a hardening recommendation related to the same service) and which, if reported by an agent, are neither counted as True Positives nor as False Positives. This avoids penalising agents for legitimate adjacent findings while keeping the recall denominator fixed. The evaluator leniency this introduces is discussed in §8 (Limitations).

4.5. Evaluator

The evaluator (§ 5.7 integrates results into Phase 6) matches each LLM finding against ground truth in three ordered passes, with each pass consumed at most once per GT entry: (1) exact CVE match; (2) `(ip, vuln_type)` with type canonicalised through the taxonomy (§5.8); (3) a last-resort `(ip, severity)` fallback for findings whose type string is unknown but whose affected host and impact match a GT entry. Findings that match none of the three passes are counted as False Positives. Severity weights are $w = \{\text{critical=4}, \text{high=3}, \text{medium=2}, \text{low=1}\}$; loose fallback matches (pass 3) apply $0.5\times$, severity mismatches within an otherwise valid match apply $0.75\times$. The composite score is the weighted True-Positive sum normalized by $\sum_{g\in\text{GT}} w(g)$.

Severity-aware dedup. Before evaluation, the Phase 3 aggregation step canonicalises vulnerability types, removes known noise classes, and deduplicates collisions on `(ip, type, port)` by retaining the *lower* reported severity. This conservative rule counteracts the severity inflation we observed in pilot runs and prevents duplicate findings from increasing match scores.

Evidence levels. Beyond match accuracy, the evaluator tracks, per True Positive, the deepest *evidence level* achieved by the harness: L1 (detection, banner or port), L2 (exploitation, authenticated action), L3 (data exfiltration, sensitive content retrieved). We report Lk-coverage $= |\{t \in \text{TP} : \text{level}(t) \geq k\}|/|\text{TP}|$. This metric cannot be computed on single-host binary-flag benchmarks and captures a dimension of pentest quality that existing LLM-pentest evaluations conflate.

Hop depth and Multi-Hop Reach. Let $G = (V, E)$ be the firewall-permitted reachability graph (edges are firewall-allowed protocol flows from §5.2) and let $e \in V$ be the attacker entry point (the subnet ingress). For each ground-truth vulnerability g attached to device $d_g \in V$, we define its *hop depth* as the number of required pivots or zone transitions from the entry point. Directly reachable vulnerabilities have hop_depth $= 0$:

$$\text{hop\_depth}(g) \;=\; \min_{p\,\in\,\text{paths}(e\to d_g)} \max(0, |p| - 1).$$

A flat scenario has hop_depth$(g) = 0$ for every directly reachable device; a gateway-centric scenario places backend services at hop_depth ≥ 1; an IT/OT-segmented scenario places OT services at hop_depth ≥ 2. We then define the Multi-Hop Reach at depth k as the recall restricted to deep ground-truth findings:

$$\text{MHR}_k \;=\; \frac{|\{g \in \text{GT} : \text{hop\_depth}(g) \geq k \,\wedge\, g \in \text{TP}\}|}{|\{g \in \text{GT} : \text{hop\_depth}(g) \geq k\}|}.$$

Raw recall is reported separately as Task Completion; MHR_1, MHR_2, and MHR_3 measure progressively deeper behind-pivot findings. A single-host agent run on only directly exposed hosts is expected to collapse on MHR_k for $k \geq 1$ unless it can establish and use a new vantage point.

5. MeshScout: A Network-Native LLM Pentest Harness

5.1. Overview

The harness decomposes network pentest into six phases with typed deliverables. The key design decision is *parallelization at the device and vulnerability granularity*: rather than a single monolithic agent that must hold the full network state, we spawn dedicated micro-agents per target with constrained tool sets, then aggregate deterministically. This keeps each micro-agent's context bounded *per device* at $O(1)$ in the number of network devices, with cross-device state only re-introduced in Phase 5 (intrusion) and Phase 6 (report).

5.2. Phase 1: Topology prior

Phase 1 loads a graph model of the target network as a directed graph $G = (V, E)$ where V are devices and E are links labeled by protocol. Two modes are supported: (i) *scenario mode* loads the predefined topology from the benchmark YAML; (ii) *discovery mode* starts from an empty graph and instructs Phase 2 to populate V via network scans. The output is a markdown deliverable enumerating attack surface and preliminary pivot candidates (betweenness-centrality ranked).

5.3. Phase 2: Full-network discovery

Phase 2 executes active reconnaissance across the target `/24`: `nmap_discovery` (host discovery), `nmap` (service/version), `arp_scan` (L2 + vendor), and protocol-specific probes (`mqtt_listen` subscribing to `#` on discovered brokers, `ssh_audit` against SSH servers, `curl_headers` on HTTP). Unlike single-host harnesses, the agent is not given a target IP: it must derive one from the CIDR scope.

5.4. Phase 3: Per-device parallel triage

Phase 3 runs in three sub-phases:

1) **Deterministic scanner pass.** Ansible-driven subprocess wrappers (`ssh_audit`, `curl_headers`, `mqtt_listen`, `modbus_scan`, ...) run in parallel per discovered device and dump structured scan results to disk.
2) **Per-device LLM triage.** For each device, a dedicated LLM agent receives only that device's Phase 2+3a outputs and has access to a minimal tool set: `http_get` (follow-up file retrieval) and `cve_search` (CPE-to-CVE lookup via NVD). The agent produces a JSON vuln list scoped to that device.
3) **Deterministic merge.** Per-device outputs are concatenated, canonicalized through the taxonomy (§5.8), and deduplicated via severity-aware rules.

This decomposition is essential for network scale: a monolithic agent on an 8-device scenario already needs to fit ∼15–40k tokens of scan output into a single context and reason about cross-device relationships simultaneously, and the cost grows linearly with n. Per-device micro-agents flatten this to a constant size, at the cost of losing some cross-device inference capability, which is recovered in Phase 5.

5.5. Phase 4: Per-vulnerability verification with evidence levels

Phase 4 spawns one micro-agent per Phase 3 finding (in parallel, with a concurrency cap). Each micro-agent receives the finding (IP, type, port, details) and full operational tool access: `ssh_login`, `mysql_query`, `mqtt_listen`, `http_get`, `ftp_list`, `modbus_scan`, `telnet_connect`. The agent is prompted to produce the deepest evidence level it can (L1–L3, §4.5) within a bounded turn budget. Outputs include: `status` (CONFIRMED, NOT_EXPLOITABLE, ERROR), `evidence_level`, `command` executed, `stdout` excerpts, and `data_extracted` (when L3).

Findings in CONFIRMED state from Phase 3 override Phase 4 ERROR/FAILED statuses (rationale: Phase 3 is hypothesis generation with directly-observed indicators; Phase 4 crashes should not erase directly-observed findings).

5.6. Phase 5: Intrusion and lateral movement

Phase 5 turns the per-device vulnerability inventory from Phases 3–4 into a network-scoped infiltration campaign. Given the infrastructure graph (Phase 1) and the confirmed credentials/exposed services collected so far, an intrusion agent attempts: (i) *credential spraying* across discovered hosts (reusing harvested SSH/HTTP/MQTT/DB credentials), (ii) *lateral movement* along firewall-permitted edges, with each successful hop adding a new vantage point to the graph, and (iii) *crown-jewel access* on back-end services (databases, NVRs, historian, cloud APIs) that are unreachable from the entry point in segmented patterns (§4.2). The agent records, for every hop, the source node, destination node, edge protocol, and credentials/exploit used. This phase is the sole producer of L_3 evidence and of MHR_k findings at $k \geq 2$; ablating it (Tab. 5, 6) forces those columns to zero by construction.

5.7. Phase 6: Network-level report

Phase 6 consumes all prior deliverables and produces a network-scoped report with: device inventory, per-device vulnerabilities with severity, attack surface summary, multi-hop intrusion narrative (from Phase 5), and actionable recommendations ordered by severity × exposure. This is the only phase with cross-device aggregation reasoning, deliberately placed at the end to avoid bloating earlier phases.

5.8. Taxonomy and tool abstraction

A single module resolves LLM-produced vulnerability type strings into a canonical taxonomy of 11 categories

Fig. 2 placeholder: 6-phase pipeline diagram.
Phase 1 (topology prior) → Phase 2 (discovery) → Phase 3 (per-device triage) → Phase 4 (per-vuln verification) → Phase 5 (intrusion) → Phase 6 (network report).

Figure 3. MESHSCOUT pipeline. Solid arrows are deliverables; dashed boxes indicate LLM agent calls; Phase 3 and Phase 4 spawn micro-agents in parallel (§5.4–§5.5).

Fig. 3 placeholder: Phase 3 parallelization — n devices → n LLM agents in parallel → deterministic merge.

Figure 4. Per-device parallelization. Each micro-agent receives only its device's scan output, avoiding the context explosion of a monolithic network-level agent.

(§4), filtering noise (e.g., meta-observations) and CONFIG-only findings. Tools are defined declaratively as YAML (schema: name, description, JSON-schema parameters, subprocess command template), so adding a new protocol probe requires no Python code.

5.9. Complexity and cost envelope

For a network with n devices and an average of v findings per device, the harness issues $1\,(\text{P1}) + 1\,(\text{P2}) + n\,(\text{P3b}) + nv\,(\text{P4}) + 1\,(\text{P5}) + 1\,(\text{P6}) = O(nv)$ LLM agent invocations, each bounded by a fixed turn budget (50 for Phase 1, 50 for Phase 2, 10 per device for Phase 3b, 10 per finding for Phase 4, 30 for Phase 5, 20 for Phase 6). One *call* is one round of (*prompt → tool calls → response*); one *turn* is one such round inside the budget. Total cost scales linearly with network size, not combinatorially.

6. Experimental Setup

Infrastructure. All scenarios run on a single Proxmox VE host. Networks are Linux bridges (`vmbr1`, `192.168.100.0/24`) with an OpenWrt router as gateway. Per-scenario deployment and teardown are automated via Ansible and complete in under 10 minutes per scenario.

LLM model: fairness lock. To isolate *architectural* contributions from model-specific advantages, all LLM-driven systems compared in this work, including our pipeline (and its ablations) and all LLM baselines, run on the same general-purpose model: **MiniMax-M2.7** [38]. We deliberately avoid a multi-model matrix: a fair comparison of architectures requires the model variable to be held constant. CAI's specialised `alias1` model (security-fine-tuned, paywalled) is intentionally not used: this study addresses the architecture axis, not the model fine-tuning axis. The choice of MiniMax-M2.7 also keeps out-of-pocket cost bounded under our research subscription, enabling the full k=3-run variance estimation across systems and scenarios without budget trade-offs.

Baselines. The core evaluation compares MESHSCOUT against two internal ablations and one non-LLM scanner baseline. External LLM pentest frameworks are discussed as related work; we only include them in quantitative tables when their tool interface can be adapted to the same model, scenario scope, and turn budget without changing their architecture.

- **MESHSCOUT–w/o-Phase 1** disables the topology prior (`--phases 2 3 4 5 6`), testing the value of infrastructure-graph guidance.
- **MESHSCOUT–w/o-Phase 5** disables the intrusion phase (`--phases 1 2 3 4 6`), testing the value of structured multi-hop lateral movement.
- **Nmap NSE** runs Nmap with all NSE scripts (`--script=default,vuln,auth`) against the same target CIDR. Scanner outputs are converted to the evaluator's JSON schema.

Internal ablations. The two ablations above isolate our key design choices: topology-guided planning and a dedicated lateral-movement phase. Phase 5 is the only phase that intentionally establishes new vantage points, so removing it should primarily affect L_3 evidence and MHR_k for $k \geq 1$ (§5.6).

Variance. Each (system, scenario) cell is run k=3 times. We report mean, standard deviation, and 95% bootstrap confidence intervals. Pairwise differences (our pipeline vs each baseline) are tested with Mann–Whitney U on F1 and weighted Score per scenario. *Power caveat:* with $n_1 = n_2 = 3$, the two-sided Mann–Whitney U test has a minimum reachable p-value of 0.10, so individual per-scenario comparisons are indicative rather than conclusive. We mitigate this by aggregating across the ten main scenarios (combined $n = 30$ per system) for the headline cross-system tests, where power is sufficient. This protocol still exceeds the single-run reporting of PentestGPT [3], AutoPentester [7], AutoPenBench [17], and CAI [9], and matches the multi-run averaging of VulnBot [6] (5-experiment) and Pentest-R1 [10] (pass@3).

Metrics. We report five families of metrics, chosen to (i) align with prior work's expected vocabulary and (ii) surface architectural differences invisible to host-level benchmarks:

- *Task completion:* Recall against the ground-truth vulnerability set, the network-scale analogue of the sub-task / task completion rate reported by PentestGPT, VulnBot, AutoPentester, Pentest-R1, and xOffense.
- *Quality:* Precision, F1, and weighted Score (CVSS-severity-weighted Recall, §4.5). Score plays the role of AutoPentester's "vulnerability coverage" but with severity weights.
- *Multi-Hop Reach:* MHR_k for $k \in \{1, 2, 3\}$, the fraction of ground-truth vulnerabilities at `hop_depth` $\geq$

Fig. 4 placeholder: bar chart, F1 and MHR$_2$ for MESHSCOUT, *ablations, and Nmap NSE.*

Figure 5. Baseline comparison. Scanners capture known CVEs but miss configuration vulnerabilities and do not establish new vantage points. The Phase 5 ablation isolates the value of explicit lateral movement on multi-hop reach (MHR$_1$–MHR$_3$).

Fig. 5 placeholder: heatmap scenario $\times$ model, cell = weighted Score.

Figure 6. Per-scenario Score heatmap. **[TODO: Note which pattern consistently breaks LLMs.]**

k that are reached. No prior LLM-pentest benchmark measures network depth explicitly.

- *Evidence depth:* L_1 (detection), L_2 (exploitation), L_3 (exfiltration) coverage; absent from binary flag-capture benchmarks.
- *Cost envelope:* prompt+completion tokens and monetary cost (USD) from provider billing APIs, plus wall-clock time per scenario, matching CAI [9], AutoPentester, and Pentest-R1.

Compute and budget. All runs are bounded by per-phase turn caps and the concurrency limit in Phase 4. We report prompt/completion tokens, token-estimated USD cost, and actual out-of-pocket cost separately, because subscription-backed providers can make the latter lower than per-token estimates.

7. Evaluation

7.1. Overall performance

[TODO: Write §7.1 prose after final runs. Report headline F1/Score, MHR$_1$–MHR$_3$, and whether Phase 1/Phase 5 ablations explain the gap.]

7.2. Network-scale gap vs baselines

[TODO: Write §7.2 — explicit narrative: (i) scanners win on CVE/category coverage where scripts exist, (ii) LLM agents win on misconfig/default-creds, (iii) Phase 5 determines how much recall survives beyond direct reachability.]

7.3. Per-architecture analysis

[TODO: Write §7.3 — pattern-by-pattern commentary. Expected finding: gateway-centric and hub-star degrade most, because they require pivot reasoning. Edge-cloud may also degrade on cloud-side service discovery.]

7.4. Evidence-level depth

[TODO: Write §7.4 — expected: detection (L1) near-parity with scanners, but L2/L3 is LLM-only. This is the strongest argument for LLM agents in this context.]

Fig. 6 placeholder: Pareto plot, USD cost vs weighted Score, one point per system (all on MiniMax-M2.7).

Figure 7. Cost–performance Pareto across systems on MiniMax-M2.7; the multi-model frontier is deferred to future work (§10). **[TODO: Annotate which system dominates on each axis and the efficient frontier.]**

7.5. Per-protocol analysis

[TODO: Write §7.5 — 2 paragraphs. Expected: LLMs excel on MQTT/SSH/HTTP (text-rich, well-represented in training), fall short on Modbus/LoRaWAN (binary, sparse training data). Report numbers per protocol.]

7.6. Failure modes and cost/time envelope

[TODO: Write §7.6 — categorize failure modes: (a) hallucinated findings on hardened controls, (b) missed vulnerabilities on binary protocols, (c) over-reporting info-disclosure without severity calibration. Include a brief quantitative table.]

8. Discussion and Limitations

Where LLMs help, where they don't. IoTPentBench surfaces an empirical pattern: LLM agents reach high recall on misconfigurations, default credentials, and information disclosure, which are textually expressible and well represented in their training corpora. They consistently underperform on binary industrial protocols (Modbus, CoAP, LoRaWAN) and on cross-device inferences that require holding an aggregated network view. This suggests a hybrid deployment model: LLM agents for configuration auditing, traditional scanners for CVE coverage, and human operators for the OT/ICS layer.

Implications for agentic systems. The harness's per-device and per-vulnerability micro-agent structure is one answer to context scaling, but it cedes cross-device inference. Designs that keep a global aggregator agent in the loop without paying the full network-context cost are an open research direction.

Limitations. (i) *Simulated infrastructure.* IoTPentBench scenarios emulate IoT devices on Linux VMs; real constrained hardware introduces embedded-specific attack surfaces (bootloaders, RTOS, firmware) we do not cover. (ii) *Evaluator leniency.* Bonus categories are accepted as non-FP even without ground-truth matches, which inflates precision. (iii) *Model variance.* With k=3 runs per configuration, confidence intervals on F1 remain wide for some scenarios. (iv) *Prompt sensitivity.* We fix a single prompt family across models; per-model prompt tuning would likely raise scores. (v) *Static architectures.* The five patterns are fixed; real deployments evolve over time and include devices not represented here (RFID badge readers, industrial PLCs, medical devices).

TABLE 5. MAIN RESULTS: TASK COMPLETION (RECALL), PRECISION, F1, WEIGHTED SCORE, MULTI-HOP REACH (MHR_k), TOKENS, AND USD COST; AVERAGED OVER THE 10 MAIN SCENARIOS × 3 RUNS. THE 3 SCALABILITY STRESSORS AND 2 HARDENED CONTROLS ARE REPORTED SEPARATELY IN §7.3 AND §7.6. ALL LLM SYSTEMS USE `MINIMAX-M2.7` (CF. §6); NMAP NSE IS THE NON-LLM LOWER BOUND. *Task Completion* IS THE NETWORK-SCALE ANALOGUE OF THE SUB-TASK COMPLETION RATE REPORTED BY PENTESTGPT, VULNBOT, AUTOPENTESTER, AND XOFFENSE (§7.1). MHR_k MEASURES THE FRACTION OF GROUND-TRUTH VULNERABILITIES AT `HOP_DEPTH` $\geq k$ THAT WERE REACHED; "–" INDICATES NO GT ENTRIES AT THAT DEPTH. CELLS PRE-FILLED WITH 0.00 INDICATE CONFIGURATIONS THAT DO NOT ESTABLISH NEW VANTAGE POINTS AND THEREFORE CANNOT REACH DEEPER ZONES BY CONSTRUCTION.

	Quality				Multi-Hop Reach			Cost	
System	TC	Prec.	F1	Score	MHR_1	MHR_2	MHR_3	Tokens (M)	USD
MESHSCOUT (full)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MESHSCOUT w/o Phase 1 (no infra graph)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MESHSCOUT w/o Phase 5 (no intrusion)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	0.00	0.00	0.00	[TODO:]	[TODO:]
Nmap NSE (non-LLM scanner)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	0.00	0.00	0.00	–	–

TC = Task Completion (Recall against GT vuln set). Reported across k=3 runs with 95% bootstrap CI and pairwise Mann–Whitney U vs each baseline; this protocol exceeds the single-run reporting of PentestGPT, AutoPentester, AutoPenBench, and CAI, and matches the multi-run averaging of VulnBot and Pentest-R1.

TABLE 6. EVIDENCE-LEVEL COVERAGE. L_k IS THE FRACTION OF TRUE POSITIVES REACHING LEVEL k OR DEEPER. ABSENT FROM BINARY-FLAG BENCHMARKS.

System	L_1 (detect)	L_2 (exploit)	L_3 (exfiltrate)
MESHSCOUT (full)	[TODO:]	[TODO:]	[TODO:]
MESHSCOUT w/o Phase 5	[TODO:]	[TODO:]	0.00
Nmap NSE	–	–	–

9. Ethical Considerations

Isolation. All experiments run on an isolated Proxmox cluster with a dedicated bridge (`vmbr1`) and no route to external networks; the OpenWrt gateway's `WAN` interface is firewall-isolated. No live hosts, production services, or third-party infrastructure are probed.

Responsible disclosure. IoTPentBench vulnerabilities are deterministically injected into our own lab VMs via Ansible. No 0-days are introduced; all categories (misconfig, weak credentials, Terrapin CVE-2023-48795, outdated Dropbear) correspond to patterns already publicly documented. No responsible-disclosure process is triggered.

Misuse risk of released artifact. The benchmark artifact (Ansible playbooks, ground truth, evaluator) does not embed exploitation payloads for unknown vulnerabilities, and the injected weaknesses are common misconfigurations documented in OWASP IoT Top 10. The harness uses only standard pentest tools already in widespread use. We judge the net societal benefit (reproducible evaluation of LLM pentest agents, and a concrete evaluation surface for future defenders) to exceed the incremental misuse risk.

IRB / human subjects. No human subjects are involved. No personal data is collected or processed.

LLM costs and environmental footprint. We report token-estimated cost, actual API billing, and wall-clock time with the experimental results. Energy usage is not claimed unless measured from provider- or host-side telemetry.

10. Conclusion

We presented IoTPentBench, the first reproducible benchmark for evaluating LLM penetration testing agents at *network scale* on IoT infrastructure, and MESHSCOUT, a network-native multi-agent harness whose infrastructure-graph guidance and dedicated lateral movement phase keep LLM context tractable while explicitly modelling reachability. Holding the LLM constant (§6) across our pipeline and two ablations, and comparing against Nmap NSE as a non-LLM baseline, we isolate the *architectural* contribution: topology guidance and a dedicated lateral-movement phase can be measured directly through MHR_k and L1–L3 evidence coverage, dimensions that binary flag-capture benchmarks conflate.

Future work. (i) *Multi-model robustness:* extending the architectural comparison to recent frontier and open-weight models to confirm model-independence of the architectural advantage; (ii) *Real hardware:* extending to embedded Zigbee, LoRaWAN, and sub-GHz attack surfaces via SDR tooling; (iii) *Defender's side:* using IoTPentBench as a benchmark for LLM-assisted IoT defense (detection, patching recommendation).

All code, scenarios, ground truth, and run logs are released at `ANONYMIZED_URL` under an open license.

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